Extending the Powell Stochastic Optimization Framework: Generalizing Exogenous Demand Distributions via Partially Observable Competitive Latent State Dynamics

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Abstract

Warren Powell's canonical five-element framework has unified the field of sequential decision analytics (SDA), organizing stochastic optimization across 15 different fields into a single, cohesive modeling paradigm (State, Decision, Exogenous Information, Transition, Objective). Traditionally, the exogenous information process W_{t+1} representing customer demands or "loads" is modeled as an independent or purely exogenous stochastic process. However, in realistic, highly competitive markets, the distribution of incoming demands depends dynamically on latent market capacity, pricing strategies, and competitor postures. In this paper, we extend the Powell framework by generalizing the exogenous load distribution via a Partially Observable Markov Decision Process (POMDP) model. By collapsing the unobservable actions of N_c competitors into a single, latent aggregate market state Θ_t, we preserve mathematical tractability and avoid the severe data-scarcity and nested-belief hurdles that render Interactive POMDPs (i-POMDPs) practically infeasible. We prove that the standard Powell model corresponds to the mathematically degenerate case where the competitor count $N_c \to 0$, collapsing the latent state simplex into a singleton and recovering the classical exogenous demand distribution. We conclude by showing how Powell's four universal classes of policies (PFAs, CFAs, VFAs, and DLAs) can be naturally deployed over the resulting belief-state MDP. This work establishes a rigorous mathematical bridge between competitive game-theoretic market dynamics and scalable, single-agent sequential decision architectures.

1 Introduction

In high-scale logistical operations, such as truckload trucking, maritime shipping, and global supply chains, decision-makers are faced with a continuous stream of sequential decisions under deep uncertainty. Shippers request transport services (referred to as "loads"), and carriers must decide in real time whether to accept or decline these requests, and which driver or asset to dispatch to each load. To solve these problems, carriers must balance immediate short-term contributions with long-term, downstream network balance and capacity constraints.

Professor Warren B. Powell has synthesized these complex, multi-stage stochastic problems under the umbrella of Sequential Decision Analytics (SDA) [1, 2]. By demonstrating that every sequential decision problem can be rigorously modeled using five core elements (State, Decision, Exogenous Information, Transition, and Objective) and solved via four universal classes of policies (PFAs, CFAs, VFAs, and DLAs), the Powell framework has provided a powerful alternative to the disjointed terminologies of reinforcement learning, optimal control, and stochastic programming.

Despite its immense success and real-world deployment (e.g., through Optimal Dynamics' CORE.ai engine, which manages thousands of driver-load pairings in real time [11, 12]), a structural limitation remains: the exogenous information process W_{t+1} representing load offers is conventionally treated as a pure exogenous process. In this standard paradigm, load arrivals are assumed to be independent of the firm's decisions and competitor actions, or governed by stationary distributions.

In a real-world market, however, the arrival and booking of loads are highly endogenous. Whether a shipper offers a load to a carrier depends on the carrier's pricing, the overall volume of freight (demand), and the capacity and pricing postures of all other competitors. To capture these interactions, one might turn to game-theoretic extensions like Interactive POMDPs (i-POMDPs) or Decentralized POMDPs (Dec-POMDPs) [6, 10]. Yet, the detailed competitor data required by i-POMDPs (such as competitors' real-time GPS locations, commitments, and exact pricing utility functions) is completely unavailable due to proprietary confidentiality and market fragmentation.

To address this challenge, we propose a competitive extension of the Powell framework that generalizes the exogenous demand distribution. Our model collapses the competitor pool into a single latent market state Θ_t within a Partially Observable Markov Decision Process (POMDP) framework [3, 4]. This approach preserves the computational tractability of single-agent sequential decision-making while explicitly modeling competitive feedback loops. We mathematically prove that when the competitor pool vanishes ($N_c \to 0$), the latent state simplex degenerates to a single point, exactly recovering the standard Powell framework.

The remainder of this paper is structured as follows. Section 2 reviews the standard Powell sequential decision framework. Section 3 introduces our latent-state POMDP formulation for generalizing exogenous loads. Section 4 discusses why this approach successfully bypasses the intractability of i-POMDPs. Section 5 presents the exact Bayesian filtering equations. Section 6 analyzes the mathematical degeneracy of the model when $N_c = 0$. Section 7 outlines policy design across Powell's four universal classes, with an expanded mathematical and algorithmic discussion of parametric CFA optimization in Section 7.2. Finally, Section 8 concludes and discusses future work.

2 The Standard Powell Sequential Decision Framework

Following Powell (2022) [1], any sequential decision problem can be represented as a sequence of states, decisions, and exogenous information:

$$(S_0, x_0, W_1, S_1, x_1, W_2, \dots, S_t, x_t, W_{t+1}, S_{t+1}, \dots) \tag{1}$$

where S_t is the state variable, x_t is the decision variable, and W_{t+1} is the exogenous information that is revealed after making decision x_t. The canonical framework is defined by five core components.

2.1 State Variables (S_t)

The state S_t represents all the information required to model the system from time t onward. It is structured into three classes:

$$S_t = (R_t, I_t, B_t) \tag{2}$$

where R_t is the resource state (e.g., the physical locations of drivers, trucks, or inventory), I_t represents the information state (e.g., current spot prices, fuel costs, weather), and B_t represents the belief state (representing probability distributions over parameters that are not known with certainty, such as customer pick-up speed or demand forecasting models [13, 14]).

2.2 Decision Variables (x_t)

The decision $x_t \in \mathcal{X}(S_t)$ is selected at time t using a policy $X^\pi(S_t)$, where $\mathcal{X}(S_t)$ is the feasible decision space. In a truckload network, x_t typically represents matching a driver $d \in \mathcal{D}_t$ to a load $\ell \in \mathcal{L}_t$, and setting the corresponding spot price p_t to charge the shipper.

2.3 Exogenous Information (W_{t+1})

The exogenous information W_{t+1} represents new information that is learned for the first time between t and $t+1$. This is the primary driver of uncertainty in the system:

$$W_{t+1} = (\hat{D}_{t+1}, \hat{p}_{t+1}, \hat{\tau}_{t+1}) \tag{3}$$

where $\hat{D}_{t+1}$ represents newly offered customer loads, $\hat{p}_{t+1}$ represents changes in market clearing prices, and $\hat{\tau}_{t+1}$ represents transit delays.

2.4 Transition Function (S^M)

The transition function $S^M(\cdot)$ is a deterministic algebraic mapping that updates the state variable based on the current state, the decision, and the incoming exogenous information:

$$S_{t+1} = S^M(S_t, x_t, W_{t+1}) \tag{4}$$

For example, if a vehicle is dispatched under decision x_t and a transit delay $\hat{\tau}_{t+1}$ is observed, its location and availability in R_{t+1} are updated accordingly.

2.5 Objective Function

Let $C_t(S_t, x_t)$ be the contribution (or cost) received at time t under state S_t and decision x_t. The objective of the decision-maker is to find a policy $\pi \in \Pi$ that maximizes the expected discounted contribution over a planning horizon T:

$$\max_{\pi \in \Pi} \mathbb{E}\left[\sum_{t=0}^{T} \gamma^t C_t(S_t, X^\pi(S_t)) \,\middle|\, S_0 \right] \tag{5}$$

subject to the state transition dynamics and a known probability model for the exogenous information sequence $W_1, W_2, \ldots, W_T$.

In standard carrier models, the exogenous load arrivals $\hat{D}_{t+1}$ are assumed to follow a stochastic process $P(W_{t+1} \mid S_t)$ that is independent of the actions of other agents in the market. Under this formulation, the carrier operates as a price-taker, and market pricing or competitor interactions are either ignored or subsumed into stationary noise.

3 The POMDP Competitive Market Model

To model a competitive market where competitor decisions are private and latent, we generalize the exogenous demand distribution. We introduce a partially observable competitive market state that governs the availability and pricing of freight.

Let there be a market consisting of the focal firm (the carrier) and a set of N_c competitors. Let $\Theta_t \in \mathcal{H}$ be a high-dimensional, latent competitive market state. Θ_t represents the aggregate capacity distribution of competitors across lanes, their pricing postures, and their outstanding contractual commitments.

We formalize this competitive sequential decision problem as a Partially Observable Markov Decision Process (POMDP) defined by the tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{O}, T_\Theta, Z, R_C, \gamma \rangle$, where:

- **State Space ($\mathcal{S}$):** The true, complete state of the system is factored as $s_t = (R_t, \Theta_t) \in \mathcal{S}$. The resource state $R_t \in \mathcal{R}$ represents the focal firm's fleet status, which is fully observable. The competitive state $\Theta_t \in \mathcal{H}$ is hidden and must be inferred from observations.

- **Action Space ($\mathcal{A}$):** The firm's action $a_t = (x_t, p_t) \in \mathcal{A}$ is a joint decision consisting of physical asset allocations $x_t \in \mathcal{X}(R_t)$ and pricing/bidding decisions $p_t \in \mathcal{P}$ offered to the market.

- **Observation Space ($\mathcal{O}$):** Instead of pure, independent exogenous information, the firm receives an observation $o_t = W_t = (D_t, Y_t) \in \mathcal{O}$, where D_t is the set of realized load offers, and Y_t represents competitive market feedback (e.g., whether a bid price was accepted, spot price adjustments, or win/loss notifications).

3.1 Latent Market State Transition Dynamics

The unobserved competitive state Θ_t transitions stochastically according to a transition probability density:

$$T_\Theta(\Theta_{t+1} \mid \Theta_t, a_t) = P(\Theta_{t+1} \mid \Theta_t, x_t, p_t) \tag{6}$$

This formulation represents an endogenous transition: the firm's dispatch decisions x_t and pricing postures p_t can stochastically influence the future competitive state Θ_{t+1}. For example, aggressive pricing p_t in a specific lane might displace competitor capacity, shifting their spatial distribution in the next epoch.

3.2 The Generalized Exogenous Load Observation Model

The core contribution of this model is the reformulation of the exogenous demand distribution. Rather than load arrivals W_{t+1} being purely exogenous and independent, they are modeled as an observation emission probability distribution conditioned on the latent competitive state Θ_{t+1} and the firm's action a_t:

$$Z(o_{t+1} \mid s_{t+1}, a_t) = P(D_{t+1}, Y_{t+1} \mid R_{t+1}, \Theta_{t+1}, x_t, p_t) \tag{7}$$

Under this formulation, the probability of receiving a load offer is dynamically coupled to the competitor fleet concentration and pricing posture represented by Θ_{t+1}. If competitors have high capacity in a specific lane, the probability of the firm receiving a load offer D_{t+1} decreases. Similarly, the probability of winning a bid Y_{t+1} depends on the carrier's bid p_t relative to the unobserved price distributions of its competitors.

The observable resource state R_t continues to transition deterministically given the observable allocation decision x_t and the realized loads D_{t+1}:

$$R_{t+1} = f_R(R_t, x_t, D_{t+1}) \tag{8}$$

This architecture preserves the physical structure of Powell's transition function while embedding the competitive market feedback loop within the observation model.

4 Bypassing Interactive POMDPs via Competitor Collapsing

An alternative approach to modeling competitive markets is the Interactive Partially Observable Markov Decision Process (i-POMDP). In an i-POMDP, each competitor $i \in \{1, \ldots, N_c\}$ is modeled as an individual agent with its own private physical state R_t^i, actions a_t^i, and private beliefs b_t^i about the market and about the focal firm's beliefs.

This model requires the focal firm to maintain a nested belief state of the form:

$$b_t^0 = \left(R_t, P\left(\Theta_t, R_t^1, b_t^1, R_t^2, b_t^2, \ldots, R_t^{N_c}, b_t^{N_c}\right)\right) \tag{9}$$

Solving an i-POMDP requires tracking infinite-dimensional hierarchies of beliefs (beliefs about beliefs about beliefs), which is computationally intractable even for very small toy problems. Furthermore, from a data-engineering perspective, the i-POMDP model is impossible to deploy in real-world logistics. Carriers do not have access to competitors' fleet capacity distributions, load schedules, or pricing models.

By collapsing the N_c competitors into a single latent market state Θ_t, our POMDP model bypasses both the computational and data-scarcity bottlenecks:

$$\Theta_t = g\left(R_t^1, a_t^1, R_t^2, a_t^2, \ldots, R_t^{N_c}, a_t^{N_c}\right) \tag{10}$$

where $g(\cdot)$ is an aggregation function that maps the joint competitor configurations into an equivalent aggregate market capacity and pricing posture.

This collapse results in a standard, single-agent POMDP over the expanded state space $\mathcal{R} \times \mathcal{H}$. Rather than tracking individual competitor beliefs, the focal firm only needs to maintain a belief $b_t(\Theta_t)$ over the aggregate, low-dimensional sufficient statistic of the market. This belief can be tractably updated using observable, historical transaction data, such as load booking rates and win/loss feedback.

5 Belief Filtering in the Extended Framework

Because the competitive state Θ_t is latent, the focal carrier cannot make decisions based on the true state (R_t, Θ_t). Instead, the carrier must maintain a belief state $b_t \in \Delta(\mathcal{H})$, where $\Delta(\mathcal{H})$ is the simplex of probability distributions over the hidden market state space $\mathcal{H}$.

The belief state $b_t(\Theta_t) = P(\Theta_t \mid H_t)$ represents the probability that the market is in competitive state Θ_t given the history of observed load offers, bid outcomes, and pricing decisions up to time t:

$$H_t = (b_0, a_0, o_1, a_1, o_2, \ldots, a_{t-1}, o_t) \tag{11}$$

When the firm executes action $a_t = (x_t, p_t)$ and subsequently receives the exogenous observation $o_{t+1} = W_{t+1} = (D_{t+1}, Y_{t+1})$, the belief state is updated recursively using Bayes' rule.

Theorem 1 (Recursive Competitive Belief Update)**.** *Given the current belief state $b_t(\Theta_t)$, the action $a_t = (x_t, p_t)$, the updated resource state $R_{t+1} = f_R(R_t, x_t, D_{t+1})$, and the exogenous observation $W_{t+1} = (D_{t+1}, Y_{t+1})$, the updated belief state $b_{t+1}(\Theta_{t+1})$ is given by:*

$$b_{t+1}(\Theta_{t+1}) = \frac{Z\left((D_{t+1}, Y_{t+1}) \mid R_{t+1}, \Theta_{t+1}, a_t\right) \sum_{\Theta_t \in \mathcal{H}} T_\Theta(\Theta_{t+1} \mid \Theta_t, a_t) b_t(\Theta_t)}{P(D_{t+1}, Y_{t+1} \mid R_{t+1}, b_t, a_t)} \tag{12}$$

where the normalizing denominator is the marginal probability of the observation:

$$P(D_{t+1}, Y_{t+1} \mid R_{t+1}, b_t, a_t) = \sum_{\Theta'_{t+1}} Z\left((D_{t+1}, Y_{t+1}) \mid R_{t+1}, \Theta'_{t+1}, a_t\right) \sum_{\Theta'_t} T_\Theta(\Theta'_{t+1} \mid \Theta'_t, a_t) b_t(\Theta'_t) \tag{13}$$

Proof. By definition of the belief state and the Markov properties of the POMDP:

$$\begin{aligned} b_{t+1}(\Theta_{t+1}) &= P(\Theta_{t+1} \mid H_{t+1}) \\ &= P(\Theta_{t+1} \mid H_t, a_t, R_{t+1}, D_{t+1}, Y_{t+1}) \\ &= \frac{P(D_{t+1}, Y_{t+1} \mid R_{t+1}, \Theta_{t+1}, H_t, a_t) P(\Theta_{t+1} \mid H_t, a_t)}{P(D_{t+1}, Y_{t+1} \mid R_{t+1}, H_t, a_t)} \end{aligned}$$

Using the observation model's conditional independence (eq. 7), the first term in the numerator simplifies to:

$$P(D_{t+1}, Y_{t+1} \mid R_{t+1}, \Theta_{t+1}, H_t, a_t) = Z\left((D_{t+1}, Y_{t+1}) \mid R_{t+1}, \Theta_{t+1}, a_t\right)$$

The second term in the numerator is the predictive distribution of the latent state, which expands by marginalizing over the previous hidden state Θ_t:

$$\begin{aligned} P(\Theta_{t+1} \mid H_t, a_t) &= \sum_{\Theta_t \in \mathcal{H}} P(\Theta_{t+1} \mid \Theta_t, H_t, a_t) P(\Theta_t \mid H_t, a_t) \\ &= \sum_{\Theta_t \in \mathcal{H}} T_\Theta(\Theta_{t+1} \mid \Theta_t, a_t) b_t(\Theta_t) \end{aligned}$$

Substituting these expressions back into the Bayes' relation, and expanding the denominator via the law of total probability over all possible future competitive states Θ'_{t+1}, yields equations (12) and the normalizing denominator, completing the proof. □

By maintaining this belief state b_t, the focal carrier can convert the partially observable sequential decision problem into a fully observable belief-state Markov Decision Process (belief MDP). The expanded state variable in the Powell framework is now written as:

$$S_t^{ext} = (R_t, I_t, b_t) \tag{14}$$

where $b_t \in \Delta(\mathcal{H})$ represents the belief state component B_t of the standard Powell state vector.

6 Mathematical Degeneracy to the Powell Framework ($N_c \to 0$)

We now demonstrate that the standard Powell sequential decision framework is a mathematically "degenerate" case of our competitive POMDP formulation. This occurs when the number of market competitors N_c goes to zero, corresponding to a carrier operating in a monopolistic or isolated market.

Theorem 2 (Market Collapse to the Powell Degenerate Case)**.** *As the number of competitors $N_c \to 0$, the latent market state space $\mathcal{H}$ collapses to a singleton $\{\Theta_\emptyset\}$. Under this collapse, the recursive belief update (eq. 12) degenerates to a static point mass, and the generalized observation model (eq. 7) decouples from latent dynamics, exactly recovering the standard, purely exogenous Powell framework.*

Proof. Let $N_c = 0$. Since there are no competitors in the market, there are no hidden competitor capacity distributions, commitments, or pricing strategies to track. The latent market state space $\mathcal{H}$ consists of a single state representing competitor absence:

$$\mathcal{H} = \{\Theta_\emptyset\} \tag{15}$$

The competitive state transition function T_Θ must satisfy:

$$T_\Theta(\Theta_\emptyset \mid \Theta_\emptyset, a_t) = P(\Theta_{t+1} = \Theta_\emptyset \mid \Theta_t = \Theta_\emptyset, a_t) = 1, \quad \forall a_t \in \mathcal{A} \tag{16}$$

Similarly, the firm's belief state $b_t \in \Delta(\mathcal{H})$ over time must be a Dirac delta distribution centered at $\Theta_\emptyset$:

$$b_t(\Theta_t) = \delta(\Theta_t - \Theta_\emptyset) = \begin{cases} 1 & \text{if } \Theta_t = \Theta_\emptyset \\ 0 & \text{otherwise} \end{cases} \tag{17}$$

We evaluate the recursive belief update for $b_{t+1}(\Theta_{t+1})$ under these conditions. From equation (12):

$$\begin{aligned} b_{t+1}(\Theta_\emptyset) &= \frac{Z\left((D_{t+1}, Y_{t+1}) \mid R_{t+1}, \Theta_\emptyset, a_t\right) T_\Theta(\Theta_\emptyset \mid \Theta_\emptyset, a_t) b_t(\Theta_\emptyset)}{Z\left((D_{t+1}, Y_{t+1}) \mid R_{t+1}, \Theta_\emptyset, a_t\right) T_\Theta(\Theta_\emptyset \mid \Theta_\emptyset, a_t) b_t(\Theta_\emptyset)} \\ &= \frac{Z\left((D_{t+1}, Y_{t+1}) \mid R_{t+1}, \Theta_\emptyset, a_t\right) \cdot 1 \cdot 1}{Z\left((D_{t+1}, Y_{t+1}) \mid R_{t+1}, \Theta_\emptyset, a_t\right) \cdot 1 \cdot 1} = 1 \end{aligned}$$

Thus, the belief update is static and redundant, providing zero new information. The belief state remains $b_t = \delta(\Theta_t - \Theta_\emptyset)$ for all t.

Next, we examine the generalized exogenous load observation model. Under the collapse of $\mathcal{H}$, the joint observation probability Z simplifies to:

$$\begin{aligned} Z(o_{t+1} \mid s_{t+1}, a_t) &= P(D_{t+1}, Y_{t+1} \mid R_{t+1}, \Theta_{t+1} = \Theta_\emptyset, x_t, p_t) \\ &= P(D_{t+1}, Y_{t+1} \mid R_{t+1}, x_t, p_t) \end{aligned}$$

Because $N_c = 0$, competitive feedback is absent. The bidding feedback variable Y_{t+1} (which represents competitive win/loss signals) collapses, leaving only the physical load offers D_{t+1}. The pricing decision p_t simplifies to a contract price that is accepted or rejected based solely on the shipper's stationary demand curve. The load offers D_{t+1} transition back to a purely exogenous process governed by market-wide parameters in the information state I_{t+1}:

$$P(D_{t+1} \mid R_{t+1}, x_t, p_t) = P(D_{t+1} \mid I_{t+1}) = P(W_{t+1} \mid S_t) \tag{18}$$

The expanded state variable S_t^{ext} collapses:

$$S_t^{ext} = (R_t, I_t, b_t = \delta(\Theta_t - \Theta_\emptyset)) \equiv (R_t, I_t) = S_t \tag{19}$$

This is exactly the standard, fully observable state variable of the classical Powell framework. The transition dynamics and the exogenous information process W_{t+1} are decoupled from latent competitive feedback, exactly recovering the degenerate monopolistic Powell model. □

7 Policy Design Over the Belief MDP

By converting the competitive POMDP into a fully observable belief MDP over the state $S_t^{ext} = (R_t, I_t, b_t)$, we can deploy Powell's four universal classes of policies to solve for optimal dispatch and pricing.

Table 1: Mathematical formulations of Powell's four universal policy classes extended over the competitive belief MDP.

Policy Class	Mathematical Formulation over Belief MDP
PFA	$X^{PFA}(S_t^{ext} \mid \theta) = f(R_t, I_t, b_t \mid \theta)$
CFA	$X^{CFA}(S_t^{ext} \mid \theta) = \arg\max_{x \in \mathcal{X}_t(\theta)} \bar{C}^\pi(S_t^{ext}, x \mid \theta)$
VFA	$X^{VFA}(S_t^{ext}) = \arg\max_{a_t \in \mathcal{A}_t} \left\{ C(S_t^{ext}, a_t) + \bar{V}_t^x(S_t^{ext,x}) \right\}$
DLA	$X^{DLA}(S_t^{ext}) = \arg\max_{a_t \in \mathcal{A}_t} \left\{ C(S_t^{ext}, a_t) + \tilde{\mathbb{E}} \left[\max_{\tilde{\pi}} \sum_{t'=t+1}^{t+H} \gamma^{t'-t} C_{t'}(\tilde{S}_{t'}^{ext}, X^{\tilde{\pi}}(\tilde{S}_{t'}^{ext})) \;\middle|\; S_t^{ext,x} \right] \right\}$

7.1 Policy Function Approximations (PFAs)

PFAs are analytical rules or lookup tables that map states directly to decisions without solving an optimization problem. In our extended model, a PFA can be parameterized by θ and modeled using a neural network that takes the resource state R_t and the belief vector b_t as input:

$$X^{PFA}(S_t^{ext} \mid \theta) = f(R_t, I_t, b_t \mid \theta) \tag{20}$$

PFAs are highly effective for simple, rule-based pricing decisions (e.g., "increase rates by 5% if the belief b_t indicates high competitor lane concentration").

7.2 Cost Function Approximations (CFAs) and Parameter Tuning

Cost Function Approximations (CFAs) solve a simplified, single-period (or static) optimization problem that is parametrically modified to account for future uncertainties and downstream network impacts [8, 2]. While often dismissed as heuristics by some theoretical communities, parametric CFAs are highly powerful strategies that have achieved massive success in industrial applications, particularly in high-dimensional resource allocation problems such as logistics and grid management [8, 15].

For the driver-to-load matching problem in a competitive logistics network, we construct a parameterized linear or integer programming model. Decisions are selected by solving a modified optimization problem over the decision space $\mathcal{X}(S_t^{ext})$:

$$X^{CFA}(S_t^{ext} \mid \theta) = \arg\max_{x \in \mathcal{X}_t(\theta)} \bar{C}^\pi(S_t^{ext}, x \mid \theta) \tag{21}$$

where $\bar{C}^{\pi}(S_t^{ext}, x \mid \theta)$ represents the parametrically modified objective function, and $\mathcal{X}_t(\theta)$ represents the modified constraint set. In our driver-to-load matching problem, we express this policy as:

$$X^{CFA}(S_t^{ext} \mid \theta) = \arg\max_{x \in \mathcal{X}_t} \sum_{d \in \mathcal{D}_t} \sum_{\ell \in \mathcal{L}_t} \left(c_{td\ell} + \theta_1 \bar{\tau}_{td} + \theta_2 b_t(\Theta_{high}) \right) x_{td\ell} \tag{22}$$

subject to matching feasibility and driver rest requirements. Here, $c_{td\ell}$ represents the immediate dispatch contribution (e.g., revenue minus operating costs). The parameter θ_1 scales the penalty for dispatch delay or expected empty travel time $\bar{\tau}_{td}$, biasing against matches that require drivers to move too far without a load. The parameter θ_2 scales the latent competitive state belief $b_t(\Theta_{high})$, representing the posterior probability that competitive fleet concentration is high in the target region. This term acts as an incentive or penalty (risk premium) to bias matches away from lanes where competitor saturation is high, thereby hedging against bidding failures or lost pricing power.

The structural advantage of this formulation is that it preserves the computational tractability of standard linear or integer programming, meaning we can solve high-dimensional resource allocation problems (e.g., dispatching thousands of trucks) in milliseconds, while using the parameter vector $\theta = (\theta_1, \theta_2)^T \in \Theta$ to make the policy robust to downstream competitive dynamics.

7.2.1 The Offline Parameter Tuning Problem

The major trade-off in the CFA paradigm is that we shift the computational burden from online tree search or value function estimation to an offline parameter tuning challenge. We define the performance of a policy parameterized by θ over a planning horizon T as:

$$F(\theta) = \mathbb{E}\left[\sum_{t=0}^{T} \gamma^t C_t(S_t^{ext}, X^{CFA}(S_t^{ext} \mid \theta)) \;\middle|\; S_0^{ext} \right] \tag{23}$$

Our objective is to find a parameter vector θ^* that maximizes the long-term expected discounted return:

$$\theta^* = \arg\max_{\theta \in \Theta} F(\theta) \tag{24}$$

In practice, the expectation in Equation (23) is analytically intractable due to the high-dimensional transition dynamics of the physical network and the latent competitive state Θ_t. Thus, we must evaluate the policy using a stochastic simulator (such as a carrier network simulator that models random load arrivals, pricing outcomes, and competitor transitions):

$$\hat{F}(\theta) = F(\theta) + \epsilon \tag{25}$$

where ϵ represents zero-mean observation noise resulting from Monte Carlo simulations. The optimization of Equation (24) is therefore a classical stochastic optimization problem (specifically, policy search) over a noisy black-box objective function.

7.2.2 Mathematical Challenges: Discontinuity and Non-Convexity

Tuning the parameters of a CFA policy is exceptionally challenging due to two mathematical properties of the objective function $F(\theta)$:

1. **Discontinuity and Non-Differentiability:** Because the decision $X^{CFA}(S_t^{ext} \mid \theta)$ is determined by solving an integer or linear programming model, the resulting decision vector is a discontinuous, step-like function of the parameters θ. As a result, $F(\theta)$ is non-differentiable and has zero-gradient almost everywhere, rendering traditional derivative-based methods (such as standard policy gradient methods in reinforcement learning) inapplicable.

2. **High Evaluation Costs:** Evaluating $\hat{F}(\theta)$ for a single configuration θ requires running a full multi-period stochastic simulation over the horizon T. For realistic logistical networks, a single simulation can take hours, limiting the number of total function evaluations.

To overcome these hurdles, we deploy derivative-free stochastic search (DFSS) and optimal learning methods.

7.2.3 Simultaneous Perturbation Stochastic Approximation (SPSA)

Although $F(\theta)$ is non-differentiable, we can approximate a pseudo-gradient using Simultaneous Perturbation Stochastic Approximation (SPSA) [16]. SPSA is highly efficient because it estimates the gradient using only two function evaluations, regardless of the dimension of θ.

At each iteration k, SPSA perturbs all elements of the parameter vector θ_k simultaneously using a random perturbation vector $\Delta_k \in \mathbb{R}^d$, where each component Δ_{ki} is drawn independently from a Rademacher distribution (± 1 with probability 0.5):

$$g_k(\theta_k) = \frac{\hat{F}(\theta_k + c_k \Delta_k) - \hat{F}(\theta_k - c_k \Delta_k)}{2c_k} \Delta_k^{-1} \tag{26}$$

where c_k is a decaying perturbation step size. The parameters are then updated using:

$$\theta_{k+1} = \Pi_\Theta \left[\theta_k + a_k g_k(\theta_k)\right] \tag{27}$$

where a_k is a decaying learning rate, and Π_Θ represents projection onto the feasible parameter space Θ. SPSA works well when the parameter space is moderate, but can become mired in local extrema due to the discontinuous nature of the underlying IP solver.

7.2.4 Optimal Learning and the Knowledge Gradient (KG) Policy

For highly non-smooth, non-convex landscapes where function evaluations are extremely expensive, we utilize optimal learning techniques to construct a belief model (metamodel) over the parameter space [7, 17].

We model our belief about the objective function $F(\theta)$ using a Gaussian Process (GP) prior:

$$F(\theta) \sim \mathcal{GP}\left(m(\theta), K(\theta, \theta')\right) \tag{28}$$

where $m(\theta)$ is the mean function and $K(\theta, \theta')$ is a covariance kernel (e.g., a Matérn or squared exponential kernel) that captures correlations between parameter configurations. When we evaluate a configuration θ^n in the simulator and observe $\hat{F}(\theta^n)$, we update our belief, yielding a posterior GP with mean $\mu^n(\theta)$ and covariance $K^n(\theta, \theta')$.

To determine the most informative parameter configuration to evaluate next, we deploy the **Knowledge Gradient (KG) policy for correlated beliefs** [7]. Let μ^n represent the vector of estimated values across a discretized parameter set. The Knowledge Gradient of sampling at configuration θ is defined as the expected increment in the maximum of our belief model after a single measurement:

$$\nu_n^{KG}(\theta) = \mathbb{E}_n \left[\max_i \mu_i^{n+1} - \max_j \mu_j^n \;\middle|\; \theta^n = \theta \right] \tag{29}$$

Because the beliefs are correlated, evaluating $F(\theta)$ at a single point θ updates our estimates μ^{n+1} across the entire parameter space. The expectation in Equation (29) can be computed analytically for Gaussian beliefs. The KG policy then selects the next parameter configuration to evaluate by maximizing the value of information:

$$\theta^n = \arg\max_{\theta \in \Theta} \nu_n^{KG}(\theta) \tag{30}$$

This approach is exceptionally sample-efficient, often finding near-optimal policy parameters within a handful of simulation runs. In high-dimensional parameter spaces, we can regularize this search using a Lasso-based Sparse Knowledge Gradient [18], which performs sparse feature selection to identify the small subset of parameters that have the most significant impact on network performance.

7.3 Value Function Approximations (VFAs)

VFAs use Bellman's equation but replace the intractable exact value function with an approximation $\bar{V}_{t+1}(S_{t+1}^{ext})$. Using the post-decision state concept to eliminate the expectation operator [1]:

$$X^{VFA}(S_t^{ext}) = \arg\max_{a_t \in \mathcal{A}_t} \left\{ C(S_t^{ext}, a_t) + \bar{V}_t^x(S_t^{ext,x}) \right\} \tag{31}$$

where $S_t^{ext,x} = (R_t^x, I_t^x, b_t^x)$ is the post-decision state variable, representing the state of the system immediately after executing decision a_t but before the arrival of new exogenous information W_{t+1}. The value function approximation is typically structured as a linear combination of basis functions or a deep Q-network trained using temporal difference learning [13, 14].

7.4 Direct Lookahead Approximations (DLAs)

DLAs optimize decisions by solving an approximate lookahead model over a planning horizon H:

$$X^{DLA}(S_t^{ext}) = \arg\max_{a_t \in \mathcal{A}_t} \left\{ C(S_t^{ext}, a_t) + \tilde{\mathbb{E}} \left[\max_{\tilde{\pi}} \sum_{t'=t+1}^{t+H} \gamma^{t'-t} C_{t'}(\tilde{S}_{t'}^{ext}, X^{\tilde{\pi}}(\tilde{S}_{t'}^{ext})) \;\middle|\; S_t^{ext,x} \right] \right\} \tag{32}$$

In our partially observable competitive setting, the lookahead expectation $\tilde{\mathbb{E}}$ is evaluated by sampling scenarios of latent competitive states $\Theta_{t'}$ and load arrivals $D_{t'}$ from the current belief b_t. Algorithms such as Partially Observable Monte Carlo Planning (POMCP) [3] or Determinized Sparse Partially Observable Trees (DESPOT) [5] can be deployed to efficiently search this lookahead tree online.

8 Conclusion and Discussion

In this paper, we extended Warren Powell's canonical sequential decision framework by generalizing the exogenous load distribution in logistics. Rather than modeling freight offers as a purely exogenous, independent process, we formulated a competitive Partially Observable Markov Decision

Process (POMDP). By collapsing N_c competitors into a single latent competitive market state Θ_t, we captured competitor capacity and pricing dynamics while preserving single-agent mathematical tractability, bypassing the intractable nested beliefs of i-POMDP models.

We proved that the standard Powell framework represents a mathematically degenerate case of our formulation when the number of competitors $N_c \to 0$. Under this degeneracy, the latent state space collapses, the belief distribution becomes static, and the generalized load distribution decouples from competitive dynamics, exactly recovering the classical Powell model. Finally, we demonstrated how Powell's four universal classes of policies can be naturally deployed over the resulting belief-state MDP, with a detailed emphasis on the offline derivative-free optimization of parametric Cost Function Approximations (CFAs) using SPSA and Gaussian Process Knowledge Gradient methods.

This model provides a mathematically rigorous yet practically scalable framework for carriers and logistical providers operating in competitive, highly volatile markets. By tracking aggregate market states via observable transaction data and optimizing decisions using parameterized policies, firms can automate strategic, tactical, and real-time operations, bridging the gap between game theory and scalable decision intelligence.

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